

Exercise 1 - SQL Fundamentals : Select & Filtering.

1. Selecting & Sorting

--- Q1 Retrieve all columns

```
SELECT *
FROM employees;
```

id	first_name	last_name	department	salary	hiredate	city
1	Alice	Green	IT	70000	2020-01-10	Johannesburg
2	Brian	Lee	HR	45000	2019-03-12	Cape Town
3	Cathy	Zulu	Finance	65000	2018-07-18	Durban
4	David	Mokoena	Marketing	50000	2021-11-05	Pretoria
5	Eva	Naidoo	IT	70000	2017-09-30	Johannesburg

--- Q2 Unique department

```
SELECT DISTINCT department
FROM employees;
```

department
IT
HR
Finance
Marketing

--- Q3 Names ordered by salary DESC

```
SELECT first_name,
       last_name
FROM employees
ORDER BY salary DESC;
```


first_name	last_name
Eva	Naidoo
Alice	Green
Cathy	Zulu
David	Nokoeni
Brian	Lee

--- Q4 Top 3 highest-paid employees

```
SELECT id,
       first_name,
       last_name,
       salary
```

FROM employees

ORDER BY salary DESC

LIMIT 3;

id	first_name	last_name	salary
5	Eva	Naidoo	12000
1	Alice	Green	70000
3	Cathy	Zulu	65000

2. Filtering with WHERE

--- Q5 Employees in IT

```
SELECT id,
       first_name,
       last_name,
       department
```

FROM employees

WHERE department = 'IT';

id	first_name	last_name	department
1	Alice	Green	IT
5	Eva	Naidoo	IT

---Q6 Finance with salary > 60000

```

SELECT id,
       first_name,
       last_name,
       department,
       salary
FROM employees
WHERE department = 'Finance' AND salary > 60000;

```

id	first_name	last_name	department	salary
3	Cathy	Zulu	Finance	65000

---Q7 HR or Marketing

```

SELECT id,
       first_name,
       last_name,
       department
FROM employees
WHERE department = 'HR' OR department = 'Marketing';

```

id	first_name	last_name	department
2	Brian	Lee	HR
4	David	Mokoena	Marketing

--- Q8 NOT in IT

```
SELECT id,
       first_name,
       last_name,
       department
```

FROM employees

WHERE department != 'IT';

id	first_name	last_name	department
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
4	David	Mokoena	Marketing

--- Q9 Using IN

```
SELECT id,
       first_name,
       last_name,
       department
```

FROM employees

WHERE department IN ('IT', 'HR', 'Finance');

id	first_name	last_name	department
1	Alice	Green	IT
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
5	Eva	Naidoo	IT

--- Q10 IT + salary > 65000 + Johannesburg

```
SELECT id,  
       first_name,  
       last_name,  
       department,  
       salary,  
       city
```

```
FROM employees
```

```
WHERE department = 'IT' AND salary > 65000 AND city = 'Johannesburg';
```

id	first_name	last_name	department	salary	city
1	Alice	Green	IT	70000	Johannesburg
5	Evd	Naidoo	IT	72000	Johannesburg